1) a)

$$\frac{1}{8-3}\int_3^8 (5x-2)\ \mathrm{d}x$$

$$=\frac{1}{5}\left[\frac{5}{2}x^2-2x\right]_3^8=\frac{1}{5}\left[(160-16)-\left(\frac{45}{2}-6\right)\right]=\frac{51}{2}$$

1) b)

$$\frac{1}{5-2}\int_2^5 5x^2\ \mathrm{d}x$$

$$=\frac{1}{3}\left[\frac{5}{3}x^3\right]_2^5=\frac{1}{3}\left[\frac{625}{3}-\frac{40}{3}\right]=65$$

1) c)

$$\frac{1}{16-9}\int_9^{16}\frac{1}{\sqrt{x}}\ \mathrm{d}x=\frac{1}{7}\int_9^{16}x^{-\frac{1}{2}}\ \mathrm{d}x$$

$$=\frac{1}{7}\left[2x^{\frac{1}{2}}\right]_9^{16}=\frac{1}{7}[8-6]=\frac{2}{7}$$

2)

$$\frac{1}{1.2--0.5}\int_{-0.5}^{1.2}(2x^2-1)\ \mathrm{d}x$$

$$=\frac{10}{17}\left[\frac{2}{3}x^3-x\right]_{-0.5}^{1.2}=\frac{10}{17}\left[\left(\frac{144}{125}-\frac{6}{5}\right)-\left(-\frac{1}{12}+\frac{1}{2}\right)\right]=-\frac{41}{150}=-0.273\ (3\ \text{SF})$$

3)

$$\frac{1}{k-0}\int_0^k x^2\ \mathrm{d}x$$

$$=\frac{1}{k}\left[\frac{1}{3}x^3\right]_0^k=\frac{1}{k}\left[\left(\frac{1}{3}k^3\right)-(0)\right]$$

$$=\frac{1}{3}k^2\ \text{as req.}$$

4)

$$\frac{1}{a-0}\int_0^a 8x^3\ \mathrm{d}x$$

$$=\frac{1}{a}[2x^4]_0^a=\frac{1}{a}[2a^4-0]=2a^3$$

5)

$$\frac{1}{b-a}\int_a^b (mx+c)\ \mathrm{d}x$$
$$= \tfrac{1}{b-a}\left[\tfrac{1}{2}mx^2+cx\right]_a^b = \tfrac{1}{b-a}\left[\left(\tfrac{1}{2}mb^2+bc\right)-\left(\tfrac{1}{2}ma^2+ca\right)\right]$$
$$= \tfrac{1}{b-a}\left[\tfrac{1}{2}m(b^2-a^2)+c(b-a)\right]$$
$$= \tfrac{1}{b-a}\left[\tfrac{1}{2}m(b-a)(b+a)+c(b-a)\right]$$
$$= \tfrac{1}{2}m(b+a)+c \text{ as req.}$$

6)

$$\frac{1}{2a-a}\int_a^{2a} (x+4)^2\ \mathrm{d}x = \frac{1}{a}\int_a^{2a} (x^2+8x+16)\ \mathrm{d}x$$
$$= \tfrac{1}{a}\left[\tfrac{1}{3}x^3+4x^2+16x\right]_a^{2a}$$
$$= \tfrac{1}{a}\left[\left(\tfrac{1}{3}(2a)^3+4(2a)^2+16(2a)\right)-\left(\tfrac{1}{3}a^3+4a^2+16a\right)\right]$$
$$= \tfrac{1}{a}\left(\tfrac{8}{3}a^3+16a^2+32a-\tfrac{1}{3}a^3-4a^2-16a\right)$$
$$= \tfrac{7}{3}a^2+12a+16$$

7)

$$\frac{1}{k-0}\int_0^k \frac{1+x}{\sqrt{x}}\ \mathrm{d}x = \frac{1}{k}\int_0^k \left(x^{-\frac{1}{2}}+x^{\frac{1}{2}}\right)\ \mathrm{d}x$$
$$= \tfrac{1}{k}\left[2x^{\frac{1}{2}}+\tfrac{2}{3}x^{\frac{3}{2}}\right]_0^k$$
$$= \tfrac{1}{k}\left[\left(2k^{\frac{1}{2}}+\tfrac{2}{3}k^{\frac{3}{2}}\right)-(0)\right]$$
$$= 2k^{-\frac{1}{2}}+\tfrac{2}{3}k^{\frac{1}{2}}$$
$$= \tfrac{2}{\sqrt{k}}+\tfrac{2\sqrt{k}}{3}$$

8)

$$\frac{1}{9-1}\int_1^9 \left(2\sqrt{x}+k\right)\,\mathrm{d}x = \frac{1}{8}\int_1^9 \left(2x^{\frac{1}{2}}+k\right)\,\mathrm{d}x$$

$$= \frac{1}{8}\left[\frac{4}{3}x^{\frac{3}{2}}+kx\right]_1^9$$

$$= \frac{1}{8}\left[(36+9k)-\left(\frac{4}{3}+k\right)\right]$$

$$= k+\frac{13}{3}$$

Since $k+\frac{13}{3}=\frac{22}{3}$ then $k=3$

9)

$$\frac{1}{5-2}\int_2^5 (x^2+1)\,\mathrm{d}x$$

$$= \frac{1}{3}\left[\frac{1}{3}x^3+x\right]_2^5$$

$$= \frac{1}{3}\left[\left(\frac{125}{3}+5\right)-\left(\frac{8}{3}+2\right)\right]$$

$$= 14$$

So that

$$\frac{1}{k-0}\int_0^k (x^2+1)\,\mathrm{d}x = 28$$

$$\frac{1}{k}\left[\frac{1}{3}x^3+x\right]_0^k = 28$$

$$\frac{1}{k}\left[\left(\frac{k^3}{3}+k\right)-0\right] = 28$$

$$\frac{k^2}{3}+1 = 28$$

So $k=\pm 9$ ∴since k is positive, $k=9$

10)

$$\frac{1}{k-0}\int_0^k (3x^2+12x-40)\ \mathrm{d}x$$
$$=\frac{1}{k}[x^3+6x^2-40x]_0^k$$
$$=\frac{1}{k}[(k^3+6k^2-40k)-0]$$
$$=k^2+6k-40$$
Since $k^2+6k-40=51$ then
$$k^2+6k-91=0$$
$$(k-7)(k+13)=0$$
Since $k>0$ then $k=7$

11) a)

$$\frac{1}{1+2}\int_{-2}^1 (ax-4)\ \mathrm{d}x$$
$$=\frac{1}{3}\left[\frac{1}{2}ax^2-4x\right]_{-2}^1$$
$$=\frac{1}{3}\left[\left(\frac{1}{2}a-4\right)-(2a+8)\right]$$
$$=\frac{1}{3}\left[-\frac{3}{2}a-12\right]$$
$$=-\frac{1}{2}a-4$$
And so $-\frac{1}{2}a-4=-5.5\Rightarrow a=3$

11) b)

$$\frac{1}{2p-p}\int_p^{2p} (3x-4)\ \mathrm{d}x$$
$$=\frac{1}{p}\left[\frac{3}{2}x^2-4x\right]_p^{2p}$$
$$=\frac{1}{p}\left[\left(\frac{3}{2}(2p)^2-4(2p)\right)-\left(\frac{3}{2}p^2-4p\right)\right]$$
$$=\frac{1}{p}\left[\frac{9}{2}p^2-4p\right]$$
$$=\frac{9}{2}p-4$$
And $\frac{9}{2}p-4=18.5\Rightarrow p=5$

12) a)

$$\frac{1}{3-1}\int_1^3 \frac{1}{6}\sqrt{t}\left(12-\frac{1}{t}\right)\,\mathrm{d}t = \frac{1}{2}\int_1^3 \left(2t^{\frac{1}{2}}-\frac{1}{6}t^{-\frac{1}{2}}\right)\,\mathrm{d}t$$

$$= \frac{1}{2}\left[\frac{4}{3}t^{\frac{3}{2}}-\frac{1}{3}t^{\frac{1}{2}}\right]_1^3 = \frac{1}{2}\left[\left(\frac{4}{3}\times 3^{\frac{3}{2}}-\frac{1}{3}\times 3^{\frac{1}{2}}\right)-\left(\frac{4}{3}-\frac{1}{3}\right)\right] = 2.675426\ldots = 2.68 \text{ (3 SF) as req.}$$

12) b)

$$v = 2t^{\frac{1}{2}}-\frac{1}{6}t^{-\frac{1}{2}}$$

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = t^{-\frac{1}{2}}+\frac{1}{12}t^{-\frac{3}{2}}$$

$$\frac{1}{3-1}\int_1^3 \left(t^{-\frac{1}{2}}+\frac{1}{12}t^{-\frac{3}{2}}\right)\,\mathrm{d}t$$

$$= \frac{1}{2}\left[2x^{\frac{1}{2}}-\frac{1}{6}x^{-\frac{1}{2}}\right]_1^3 = \frac{1}{2}\left[\left(2\times 3^{\frac{1}{2}}-\frac{1}{6}\times 3^{-\frac{1}{2}}\right)-\left(2-\frac{1}{6}\right)\right] = 0.7672716\ldots$$

$= 0.767$ ms^{-2} (3 SF)

13)

$$\frac{1}{3-0}\int_0^3 (-5x^3+12x^2+4x)\,\mathrm{d}x$$

$$= \frac{1}{3}\left[-\frac{5}{4}x^4+4x^3+2x^2\right]_0^3 = \frac{1}{3}\left[\left(-\frac{405}{4}+108+18\right)-0\right] = 8.25\ ^\circ\text{C}$$

14)

$$\frac{1}{12-0}\int_0^{12} (2t^3-7t^2+15t+40)\,\mathrm{d}t$$

$$= \frac{1}{12}\left[\frac{1}{2}t^4-\frac{7}{3}t^3+\frac{15}{2}t^2+40t\right]_0^{12} = \frac{1}{12}[(10368-4032+1080+480)-0]$$

$= 658$ rabbits

15)

$$\frac{1}{5-2}\int_2^5 (x^3 - 6x^2 + 9x + 4)\ \mathrm{d}x$$

$$= \frac{1}{3}\left[\frac{1}{4}x^4 - 2x^3 + \frac{9}{2}x^2 + 4x\right]_2^5$$

$$= \frac{1}{3}\left[\left(\frac{625}{4} - 250 + \frac{225}{2} + 20\right) - (4 - 16 + 18 + 8)\right] = \frac{33}{4}$$

$$\therefore k = \frac{33}{4}$$

16)

$$\frac{1}{3-1}\int_1^3 (x^3 - 3x^2 - 9x + 10)\ \mathrm{d}x$$

$$= \frac{1}{2}\left[\frac{1}{4}x^4 - x^3 - \frac{9}{2}x^2 + 10x\right]_1^3$$

$$= \frac{1}{2}\left[\left(\frac{81}{4} - 27 - \frac{81}{2} + 30\right) - \left(\frac{1}{4} - 1 - \frac{9}{2} + 10\right)\right] = -11$$

$$\therefore k = -11$$

17)

$$\frac{1}{k-2}\int_2^k (x^3-2)\ \mathrm{d}x$$

$$=\frac{1}{k-2}\left[\frac{1}{4}x^4-2x\right]_2^k$$

$$=\frac{1}{k-2}\left[\left(\frac{1}{4}k^4-2k\right)-(4-4)\right]$$

$$=\frac{1}{k-2}\left(\frac{1}{4}k^4-2k\right)$$

Since $\frac{1}{k-2}\left(\frac{1}{4}k^4-2k\right)=\frac{195}{4}$ then

$$4\left(\frac{1}{4}k^4-2k\right)=195(k-2)$$

$$k^4-8k=195k-390$$

$$k^4-203k+390=0$$

Let $\mathrm{p}(k)=k^4-203k+390$

$$\mathrm{p}(5)=625-1015+390=0$$

$$\mathrm{p}(2)=16-406+390=0$$

So $k=2$ and $k=5$ is a solution.

$$k^4-203k+390=(k-5)(k-2)(k^2+7k+39)$$

k needs to be real and >2

$\therefore k=5$

18) a)

$$\frac{1}{4-2}\int_2^4 (x^2+x-1)\ \mathrm{d}x$$

$$=\frac{1}{2}\left[\frac{1}{3}x^3+\frac{1}{2}x^2-x\right]_2^4=\frac{1}{2}\left[\left(\frac{64}{3}+8-4\right)-\left(\frac{8}{3}+2-2\right)\right]=\frac{34}{3}$$

18) b)

$25 - \frac{34}{3} = \frac{41}{3} \therefore c = \frac{41}{3}$

Alternative Method:

$$\frac{1}{4-2}\int_2^4 (x^2 + x - 1 + c)\,\mathrm{d}x = 25$$

$$\frac{1}{2}\left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - x + cx\right]_2^4 = 25$$

$$\left(\frac{76}{3} + 4c\right) - \left(\frac{8}{3} + 2c\right) = 50$$

$$\frac{68}{3} + 2c = 50$$

$$\therefore c = \frac{41}{3}$$

19) a)

A horizontal translation does not affect the mean value, but a vertical one does, so the mean value of g is given by $k - 1$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x = k$$

Since $\mathrm{g}(x) = \mathrm{f}(x-2) - 1$ the mean value of g is given by

$$\frac{1}{4}\int_{-2}^2 \mathrm{g}(x)\ \mathrm{d}x = \frac{1}{4}\int_{-2}^2 (\mathrm{f}(x-2) - 1)\ \mathrm{d}x$$

$$= \frac{1}{4}\int_{-2}^2 \mathrm{f}(x-2)\ \mathrm{d}x + \frac{1}{4}\int_{-2}^2 -1\ \mathrm{d}x$$

$$= \frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x + \frac{1}{4}\int_{-2}^2 -1\ \mathrm{d}x$$

$$= k - 1$$

19) b)

There has been a reflection of all the y-values so the mean value of g is given by $-k$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x = k$$

Since $\mathrm{g}(x) = -\mathrm{f}(x)$ the mean value of g is given by

$$\frac{1}{4}\int_2^6 \mathrm{g}(x)\ \mathrm{d}x = \frac{1}{4}\int_2^6 \big(-\mathrm{f}(x)\big)\ \mathrm{d}x$$

$$= -\frac{1}{4}\int_2^6 \mathrm{f}(x)\ \mathrm{d}x$$

$$= -k$$

19) c)

Due to symmetry in the intervals, the mean value has not changed, so the mean value of g is given by $k + 4$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x = k$$

Since $\mathrm{m}(x) = \mathrm{f}(-x)$ the mean value of g is given by

$$\frac{1}{4}\int_{-4}^0 \mathrm{m}(x)\ \mathrm{d}x = \frac{1}{4}\int_{-4}^0 \mathrm{f}(-x)\ \mathrm{d}x$$

$$= \frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x$$

$$= k$$

19) d)

The horizontal translation and symmetry mean the mean value does not change, but the vertical translation does affect the mean value. The mean value of g is given by $k + 3$

Alternative Method:

$$\frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x = k$$

Since $\mathrm{n}(x) = \mathrm{m}(x+1) + 3$ and $\mathrm{m}(x) = \mathrm{f}(-x)$, the mean value of n is given by

$$\frac{1}{4}\int_{-3}^1 \mathrm{n}(x)\ \mathrm{d}x = \frac{1}{4}\int_{-3}^1 (\mathrm{m}(x+1) + 3)\ \mathrm{d}x$$

$$= \frac{1}{4}\int_{-3}^1 \mathrm{m}(x+1)\ \mathrm{d}x + \frac{1}{4}\int_{-3}^1 3\ \mathrm{d}x$$

$$= \frac{1}{4}\int_0^4 \mathrm{f}(x)\ \mathrm{d}x + \frac{1}{4}\int_{-3}^1 3\ \mathrm{d}x$$

$$= k + 3$$

20) a)

A horizontal translation does not affect the mean value, but a vertical one does, so the mean value of g is given by $k + 4$

Alternative Method:

$$\frac{1}{5}\int_2^7 \mathrm{f}(x)\ \mathrm{d}x = k$$

Since $\mathrm{g}(x) = \mathrm{f}(x+1) + 4$ the mean value of g is given by

$$\frac{1}{5}\int_1^6 \mathrm{g}(x)\ \mathrm{d}x = \frac{1}{5}\int_1^6 (\mathrm{f}(x+1) + 4)\ \mathrm{d}x$$

$$= \frac{1}{5}\int_1^6 \mathrm{f}(x+1)\ \mathrm{d}x + \frac{1}{5}\int_1^6 4\ \mathrm{d}x$$

$$= \frac{1}{5}\int_2^7 \mathrm{f}(x)\ \mathrm{d}x + \frac{1}{5}\int_1^6 4\ \mathrm{d}x$$

$$= k + 4$$

20) b)

There has been a stretch of all the x-values so the mean value of g is given by k

Alternative Method:

$\frac{1}{5}\int_2^7 \mathrm{f}(x)\ \mathrm{d}x = k$

Since $\mathrm{g}(x) = \mathrm{f}(3x)$ the mean value of g is given by

$\frac{1}{5}\int_6^{21} \mathrm{f}(3x)\ \mathrm{d}x = \frac{1}{5}\int_2^7 \mathrm{f}(x)\ \mathrm{d}x$

$= k$

20) c)

There has been a stretch of all the y-values so the mean value of g is given by $\frac{1}{2}k$

Alternative Method:

$\frac{1}{5}\int_2^7 \mathrm{f}(x)\ \mathrm{d}x = k$

Since $\mathrm{m}(x) = \frac{1}{2}\mathrm{f}(x)$ the mean value of g is given by

$\frac{1}{5}\int_2^7 \mathrm{m}(x)\ \mathrm{d}x = \frac{1}{4}\int_2^7 \frac{1}{2}\mathrm{f}(x)\ \mathrm{d}x$

$= \frac{1}{2} \times \frac{1}{5}\int_2^7 \mathrm{f}(x)\ \mathrm{d}x$

$= \frac{1}{2}k$

21)

$[\mathrm{f}(x)]^2 = [-x(x-0.5)(x-1)]^2 = x^2(x-0.5)^2(x-1)^2$

$= x^2(x^2 - x + 0.25)(x^2 - 2x + 1)$

$= x^2(x^4 - 3x^3 + 3.25x^2 - 1.5x + 0.25)$

$= x^6 - 3x^5 + 3.25x^4 - 1.5x^3 + 0.25x^2$

$\frac{1}{0.6 - 0.1}\int_{0.1}^{0.6} (x^6 - 3x^5 + 3.25x^4 - 1.5x^3 + 0.25x^2)\ \mathrm{d}x$

$= 2\left[\frac{1}{7}x^7 - \frac{1}{2}x^6 + \frac{13}{20}x^5 - \frac{3}{8}x^4 + \frac{1}{12}x^3\right]_{0.1}^{0.6}$

$= 0.00112648$

$\sqrt{0.00112648} = 0.0336$ (3 SF)